INFORME TÉCNICO – ANÁLISIS DE RED CON NMAP

1. Objetivo

Realizar un análisis de red en un entorno local utilizando Kali Linux y Nmap para identificar hosts, puertos, servicios y evaluar la seguridad básica.

2. Alcance

Análisis realizado sobre el host 192.168.100.10 dentro de una red local privada.

3. Herramientas utilizadas

- Kali Linux
- Nmap

4. Metodología

4.1 Escaneo completo de puertos

Comando utilizado: nmap -p- 192.168.100.10

Figura 1. Escaneo completo de puertos

```
└─$ nmap -p- 192.168.100.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-29 22:10 -05
Nmap scan report for 192.168.100.10
Host is up (0.00042s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE
88/tcp    open  kerberos-sec
445/tcp   open  microsoft-ds
49477/tcp open  unknown
MAC Address: F4:34:F0:E1:89:2B (Apple)

Nmap done: 1 IP address (1 host up) scanned in 608.66 seconds
```

Análisis: Se identificaron los puertos 88, 445 y 49477.

4.2 Enumeración de servicios

Comando utilizado: nmap -sV -p 88,445,49477 192.168.100.10

Figura 2. Detección de servicios

```
└─$ nmap -sV -p 88,445,49477 192.168.100.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-29 22:31 -05
Nmap scan report for 192.168.100.10
Host is up (0.00048s latency).

PORT      STATE SERVICE       VERSION
88/tcp    open  kerberos-sec  Heimdal Kerberos (server time: 2026-03-30 03:31:45Z)
445/tcp   open  microsoft-ds?
49477/tcp open  unknown
MAC Address: F4:34:F0:E1:89:2B (Apple)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 164.82 seconds
```

Análisis: Se identificaron Kerberos, SMB y un puerto desconocido.

4.3 Escaneo agresivo

Comando utilizado: nmap -A 192.168.100.10

Figura 3. Escaneo avanzado

```
└─$ nmap -A 192.168.100.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-29 23:08 -05
Nmap scan report for 192.168.100.10
Host is up (0.0012s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE       VERSION
88/tcp  open  kerberos-sec  Heimdal Kerberos (server time: 2026-03-30 04:08:57Z)
445/tcp open  microsoft-ds?
MAC Address: F4:34:F0:E1:89:2B (Apple)
Device type: general purpose
Running: Apple macOS 11.X|12.X|13.X
OS CPE: cpe:/o:apple:mac_os_x:11 cpe:/o:apple:mac_os_x:12 cpe:/o:apple:mac_os_x:13
OS details: Apple macOS 11 (Big Sur) - 13 (Ventura) or iOS 16 (Darwin 20.6.0 - 22.4.0)
Network Distance: 1 hop

Host script results:
| smb2-security-mode:
|   3:0:2:
|     Message signing enabled and required
| smb2-time:
|   date: 2026-03-30T04:08:51
|   start_date: N/A
|_nbstat: NetBIOS name: IMAC-DE-JAVIER, NetBIOS user: <unknown>, NetBIOS MAC: f4:34:f0:e1:89:2b
(Apple)
|_clock-skew: -23s

TRACEROUTE
HOP RTT     ADDRESS
1   1.15 ms 192.168.100.10

OS and Service detection performed. Please report any incorrect results at https://nmap.org/subm
it/ .
Nmap done: 1 IP address (1 host up) scanned in 29.45 seconds
```

Análisis: Se estimó sistema operativo macOS y configuraciones adicionales.

4.4 Evaluación SMB

Comando utilizado: nmap --script smb-vuln* -p 445 192.168.100.10

Figura 4. Análisis SMB

```
└─$ nmap --script smb-vuln* -p 445 192.168.100.10
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-29 22:59 -05
Nmap scan report for 192.168.100.10
Host is up (0.00043s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: F4:34:F0:E1:89:2B (Apple)

Host script results:
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: EOF
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 5.20 seconds
```

Análisis: No se detectaron vulnerabilidades conocidas.

5. Resultados

Puertos abiertos: 88, 445, 49477.
Sistema operativo: macOS.
Sin vulnerabilidades críticas detectadas.

6. Conclusiones

El sistema presenta una configuración segura y no expone vulnerabilidades evidentes.

7. Recomendaciones

Mantener actualizaciones, restringir accesos y monitorear puertos.